

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
First Semester MCA (CBCS) Degree Examinations
 December 2024 – January 2025

22MCA103 – COMPUTER ORGANIZATION AND ARCHITECTURE

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions selecting **One** from each unit.

	Unit – I	Marks	BT*	CO*	PO*
1. a)	Express the Boolean function				
	(i) $F = A + B'C$ as a sum of minterms.				
	(ii) $F = xy + x'z$ as a product of maxterms.	10	L*3	1	1
b)	Simplify the Boolean expression $F=C(B+C)(A+B+C)$ by using Boolean algebra rules, specifically the absorption and distribution laws.	6	L3	1	1
c)	Simplify the expression using Boolean Laws and realize the same using NOR gates: $Y= AB+AC+BC'$.	4	L3	1	2
2. a)	Convert the following:				
	i) $(1764)_8$ into decimal.				
	ii) $(7F9)_{16}$ into binary.				
	iii) $(731)_{10}$ into Hexadecimal.	8	L3	1	1
	iv) $(1110101)_2$ into Octal.	3	L3	1	1
b)	Simplify the Boolean expression $Y=AB+ABC+ABC'$				
c)	Explain and demonstrate how the following basic gates can be constructed using only NAND gates and only NOR gates. Include both truth tables and circuit diagrams for each realization, clearly explaining the steps involved in the conversion.	9	L3	1	2
	Unit – II				
3. a)	Solve the following:				
	(i) Minimize the following function in SOP minimal form using K-Maps: $f = m(1, 5, 6, 11, 12, 13, 14) + d(4)$				
	(ii) Minimize the following function in POS minimal form using K-Maps: $F(A, B, C, D) = m(0, 1, 2, 3, 4, 5) + d(10, 11, 12, 13, 14, 15)$	10	L3	2	3
b)	Simplify using tabulation method: $F(A,B,C,D)=\sum m(0,1,2,4,6,8,9,11,13,15)$	10	L3	2	3
4. a)	Simplify the following expression using K maps. Also show the intermediate steps:				
	i) $F(A,B,C,D) = m(1,3,5,7,9,11,13,15)$				
	ii) Minimize the following function in POS form $F(A,B,C,D) = m(0,2,4,6,8,10) + d(12,13,14,15)$	10	L3	2	3
b)	Implement a full subtractor using 3:8 decoder.	10	L2	2	2
	Unit – III				
5. a)	Perform the restoring division for the given binary numbers $1000 \div 11$.	6	L3	3	3
b)	Explain the following:				
	(a) Synchronous Bus.				
	(b) Represent $1/8$ using IEEE standard floating-point notation	6	L2	3	2
c)	Explain with figure the design of a 4-bit carry look ahead adder.	8	L1	3	1
6. a)	Multiply 110101 and 011011 using Booth's algorithm.	5	L3	3	3
b)	Explain and realize the half Subtractor circuit.	5	L1	3	1
c)	Compute the following:				
	(i) Express -45 in 8 bit 2's complement form				
	(ii) Subtract 14 from 46 using 8 bit 2's complement arithmetic				
	(iii) Add -75 to $+26$ using 8 bit 2's complement arithmetic				
	(iv) Add -45.75 to $+87.5$ using 12 bit arithmetic	10	L3	3	2

P.T.O.

Unit – IV

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|-------|---|----|----|---|---|
| 7. a) | Illustrate the working of nested subroutine with the example program with diagram. | 5 | L2 | 4 | 2 |
| b) | List the name, assembler syntax and the addressing function for any 5 addressing modes and explain indirect addressing mode with an example. | 5 | L2 | 4 | 1 |
| c) | Explain the use of registers involved in the communication between processor and memory with a neat diagram. | 5 | L2 | 4 | 2 |
| d) | Demonstrate the working of shift and rotate instruction with example. | 5 | L2 | 4 | 2 |
| 8. a) | Draw and explain the connection between memory and processor via the respective registers. Also explain the step by step execution of an instruction. | 12 | L2 | 4 | 1 |
| b) | Explain the concept of subroutine nesting and processor stack with suitable ALP examples. | 8 | L2 | 4 | 2 |

Unit – V

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|--------|--|---|----|---|---|
| 9. a) | Differentiate between memory mapped I/O and I/O Mapped I/O. | 4 | L1 | 5 | 1 |
| b) | Design the interface circuit for keyboard to processor connection. | 5 | L1 | 5 | 2 |
| c) | How do modern operating systems manage concurrency and synchronization challenges when facilitating seamless communication and resource sharing across multiple devices? Describe any two advanced mechanisms employed to address these complexities. | 5 | L2 | 5 | 1 |
| d) | Summarize the comprehensive breakdown of the sequential phases governing the operations of a Small Computer System Interface (SCSI) bus, accompanied by an intricate timing diagram specifically, highlight the mechanisms through which the SCSI bus ensures proper handshaking, arbitration, data transfer, and termination during communication between initiators and targets, with an emphasis on the synchronization of signals and the timing nuances involved in these phases. | 6 | L2 | 5 | 1 |
| 10. a) | Define exception. Mention the types and explain them. | 4 | L2 | 5 | 1 |
| b) | Discuss the internal organization of a 2M x 8 asynchronous DRAM chip. | 8 | L1 | 5 | 1 |
| c) | Elaborate on the concept of Direct Memory Access (DMA), emphasizing its role in data transfer between memory and peripherals without CPU intervention. Additionally, provide an insightful explanation of Bus Arbitration, focusing on how it manages control of the bus in multi-master systems to prevent conflicts and ensure efficient resource allocation. | 8 | L2 | 5 | 2 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
